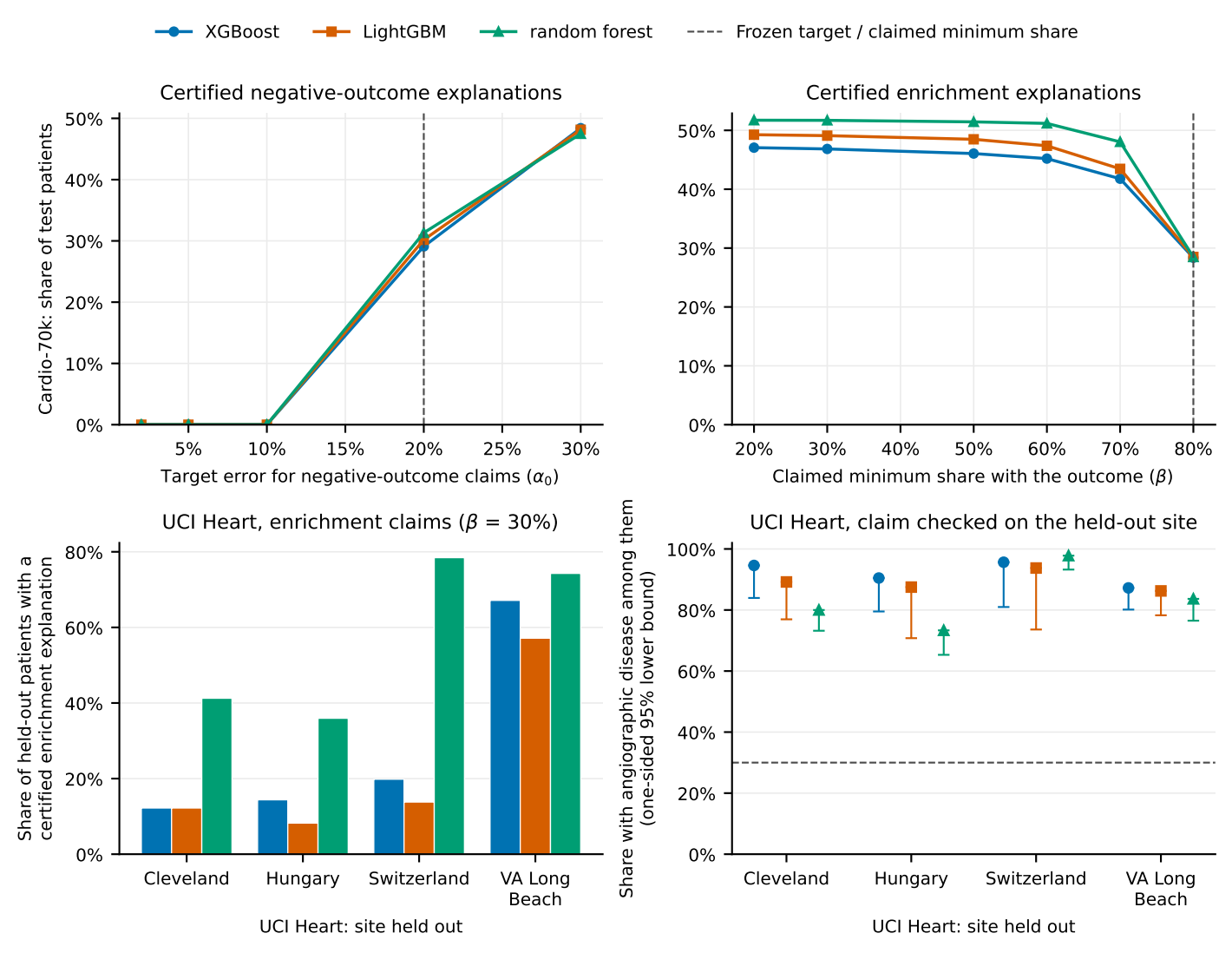
XGBoost
LightGBM
random forest
Frozen target / claimed minimum share
Certified negative-outcome explanations
Cardio-70k: share of test patients
50%
40%
30%
20%
10%
0%
5%
10%
15%
20%
25%
30%
Target error for negative-outcome claims (α_0)
Certified enrichment explanations
50%
40%
30%
20%
10%
0%
20%
30%
40%
50%
60%
70%
80%
Claimed minimum share with the outcome (β)
UCI Heart, enrichment claims (β = 30%)
Share of held-out patients with a certified enrichment explanation
80%
60%
40%
20%
0%
Cleveland
Hungary
Switzerland
VA Long Beach
UCI Heart: site held out
UCI Heart, claim checked on the held-out site
Share with angiographic disease among them (one-sided 95% lower bound)
100%
80%
60%
40%
20%
0%
Cleveland
Hungary
Switzerland
VA Long Beach
UCI Heart: site held out